

RET fusion-hotspot benchmark report

Study: msk_impact_50k_2026

Abstract

This report analyzes RET gene fusions from the msk_impact_50k_2026 study, covering 194 fusion events. 146/194 fusions (75.3%) were in-frame. 179/194 fusions (92.3%) retained the Protein kinase domain. Domain retention was tested with Fisher's exact test ($p=0.000419666$, statistically significant at $\alpha=0.05$). Compared to the literature benchmark (PMCID: PMC6430196 (KIF5B-RET): 100.0% in-frame, 100.0% domain-retained), this run observed 75.3% in-frame and 92.3% domain retention.

Results summary

Fusion partner frequency

Fusion-partner frequency was tabulated across 194 analyzed fusion events, identifying 33 distinct fusion partner genes. The most frequent partner was KIF5B, observed in 87 events.

Domain retention

Protein kinase domain retention was tested with Fisher's exact test comparing in-frame fusions against all others; 141/146 (96.6%) of in-frame fusions retained the domain, $p=0.000419666$ (statistically significant at $\alpha=0.05$). The Protein kinase domain appears to be required for retention. A breakpoint-position permutation test produced a corroborating empirical p-value of 0.000999001.

Domain disruption

Domain disruption was tested with Fisher's exact test comparing in-frame fusions against all others; 139/146 (95.2%) of in-frame fusions disrupted the domain, $p=0.00178046$ (statistically significant at $\alpha=0.05$). A breakpoint-position permutation test produced a corroborating empirical p-value of 0.635436.

Cutpoint detection

Cutpoint detection scanned 194 mapped breakpoints for the protein position that best separates domain-retained from lost/disrupted fusions; the inferred cutpoint was position 627 aa (permutation-corrected $p=0.001998$, statistically significant at $\alpha=0.05$). This is -97 aa from the nearest configured domain boundary at 724 aa.

Corroborating confidence statistics

Corroborating confidence statistics compared Domain_retention_flags groups "retained" ($n=179$) and "not_retained" ($n=15$). For Frame_status, retained: 141/157 (89.8%, 95% CI 84.1%-93.6%); not_retained: 5/14 (35.7%, 95% CI 16.3%-61.2%). Welch's t-test comparing Tumor_variant_count between groups gave means of 67.4413 vs 49.2667, $p=0.22222$ (not statistically significant at $\alpha=0.05$).

Composite evidence score

Composite evidence score produced results for this run (33 row(s) in composite_evidence_ranking); see the accompanying table.

Exon retention

Exon retention produced results for this run (1 row(s) in Exon_retention); see the accompanying table.

Joint partner

Joint partner reported: RET has no gene_pair configured; joint-partner analysis was skipped.

Tables

composite_score tables

composite_evidence_ranking

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
KIF5B	87	0.4484536082474227	0.30004340774793187	0.019692787238683573	0.8093693978698724	0.7278348081553965	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.407674382354967	1
CCDC6	51	0.26288659793814434	0.30004340774793187	0.019692787238683573	0.8167835941716136	0.7278348081553965	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.35311640870744465	2
PDCD10	1	0.005154639175257732	0.30004340774793187	0.019692787238683573	1.0	0.7278348081553965	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.30327928195283665	3
TFG	1	0.005154639175257732	0.30004340774793187	0.019692787238683573	1.0	0.7278348081553965	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.30327928195283665	4
ERC1	1	0.005154639175257732	0.30004340774793187	0.019692787238683573	0.9724770642201834	0.7278348081553965	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.29915084158586414	5
NCOA4	16	0.08247422680412371	0.30004340774793187	0.019692787238683573	0.8031823394495413	0.7278348081553965	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.2969525091589276	6
RBPM5	1	0.005154639175257732	0.30004340774793187	0.019692787238683573	0.9311926605504587	0.7278348081553965	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.2929581810354055	7
RUFY1	2	0.010309278350515464	0.30004340774793187	0.019692787238683573	0.9174311926605505	0.7278348081553965	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.2924403526044965	8
LIN52	1	0.005154639175257732	0.30004340774793187	0.019692787238683573	0.908256880733945	0.7278348081553965	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.2895178140629284	9

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
EML4	1	0.005154639175257732	0.30004340774793187	0.019692787238683573	0.8944954128440367	0.7278348081553965	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.28745359387944214	10
DOCK1	1	0.005154639175257732	0.30004340774793187	0.019692787238683573	0.8371559633027523	0.7278348081553965	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.2788526764482495	11
SPECC1L	4	0.020618556701030927	0.30004340774793187	0.019692787238683573	0.8044724770642202	0.7278348081553965	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.2785893287702016	12
TIMM23B	3	0.015463917525773196	0.30004340774793187	0.019692787238683573	0.8035168195718654	0.7278348081553965	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.27689958839377105	13
KCNMA1	1	0.005154639175257732	0.30004340774793187	0.019692787238683573	0.823394495412844	0.7278348081553965	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.2767884562647632	14
ERCC6	3	0.015463917525773196	0.30004340774793187	0.019692787238683573	0.8027522935779816	0.7278348081553965	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.2767849094946885	15
RUFY3	2	0.010309278350515464	0.30004340774793187	0.019692787238683573	0.805045871559633	0.7278348081553965	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.2755825544393589	16
CCDC186	1	0.005154639175257732	0.30004340774793187	0.019692787238683573	0.805045871559633	0.7278348081553965	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.2740361626867816	17
CUBN	1	0.005154639175257732	0.30004340774793187	0.019692787238683573	0.805045871559633	0.7278348081553965	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.2740361626867816	18
GEMIN5	1	0.005154639175257732	0.30004340774793187	0.019692787238683573	0.805045871559633	0.7278348081553965	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.2740361626867816	19

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
KIF13A	1	0.005154639175257732	0.30004340774793187	0.019692787238683573	0.805045871559633	0.7278348081553965	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.2740361626867816	20
RELCH	1	0.005154639175257732	0.30004340774793187	0.019692787238683573	0.805045871559633	0.7278348081553965	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.2740361626867816	21
SLC4A4	1	0.005154639175257732	0.30004340774793187	0.019692787238683573	0.805045871559633	0.7278348081553965	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.2740361626867816	22
CLIP1	1	0.005154639175257732	0.30004340774793187	0.019692787238683573	0.8027522935779816	0.7278348081553965	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.27369212598953385	23
GRIPAP1	1	0.005154639175257732	0.30004340774793187	0.019692787238683573	0.8027522935779816	0.7278348081553965	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.27369212598953385	24
RASSF4	1	0.005154639175257732	0.30004340774793187	0.019692787238683573	0.8027522935779816	0.7278348081553965	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.27369212598953385	25
RUFY2	1	0.005154639175257732	0.30004340774793187	0.019692787238683573	0.8027522935779816	0.7278348081553965	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.27369212598953385	26
SHROOM3	1	0.005154639175257732	0.30004340774793187	0.019692787238683573	0.8027522935779816	0.7278348081553965	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.27369212598953385	27
TRIM24	1	0.005154639175257732	0.30004340774793187	0.019692787238683573	0.8027522935779816	0.7278348081553965	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.27369212598953385	28
KIAA1217	1	0.005154639175257732	0.30004340774793187	0.019692787238683573	0.536697247706422	0.7278348081553965	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.23378386910879995	29

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
CTNNA3	1	0.005154639175257732	0.30004340774793187	0.019692787238683573	0.29587155963302747	0.7278348081553965	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.19766001589779075	30
CTNNA1	1	0.005154639175257732	0.30004340774793187	0.019692787238683573	0.1009174311926605	0.7278348081553965	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.16841689663173573	31
UXS1	1	0.005154639175257732	0.30004340774793187	0.019692787238683573	0.06422018348623848	0.7278348081553965	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.16291230947577243	32
FBXL7	2	0.010309278350515464	0.30004340774793187	0.019692787238683573	0.0	0.7278348081553965	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.15482567370541397	33

cutpoint_detection tables

cutpoint_scan

cutpoint	n_positive_at_or_below	n_positive_above	n_negative_at_or_below	n_negative_above	odds_ratio	p_value	neg_log10_p_value
219	0	179	1	14	0.0	0.07731958762886598	1.1117104708745449
425	1	178	1	14	0.07865168539325842	0.14903050050745154	0.826724840000421
550	3	176	2	13	0.11079545454545454	0.048844025264081076	1.3111885527954676
556	4	175	2	13	0.14857142857142858	0.06998082470265705	1.1550209437950567
587	5	174	3	12	0.11494252873563218	0.016699929087495975	1.7772853729826312
622	6	173	3	12	0.13872832369942195	0.023865412628667555	1.6222310523597865
627	8	171	5	10	0.0935672514619883	0.001234261100135449	2.9085929583309174
639	9	170	5	10	0.10588235294117647	0.0018331517060542683	2.7368015926613842
640	10	169	5	10	0.11834319526627218	0.0026250181738802633	2.5808676854814183
657	11	168	5	10	0.13095238095238096	0.003644511061609513	2.438360727287861
663	13	166	5	10	0.1566265060240964	0.006510395780065948	2.1863926089934114
671	15	164	5	10	0.18292682926829268	0.01072245099455107	1.9697059299659876
673	16	163	5	10	0.19631901840490798	0.01342338349072885	1.8721380023380016
687	17	162	5	10	0.20987654320987653	0.01656697136370287	1.7807568784395744
708	18	161	5	10	0.2236024844720497	0.020185752752359003	1.694955050558838
712	88	91	8	7	0.8461538461538461	0.7940441928844832	0.10015532604674418

cutpoint	n_positive_at_or_below	n_positive_above	n_negative_at_or_below	n_negative_above	odds_ratio	p_value	neg_log10_p_value
713	176	3	14	1	4.190476190476191	0.27712145920070164	0.5573298428514589
934	176	3	15	0	0.0	1.0	-0.0
1019	177	2	15	0	0.0	1.0	-0.0

domain_disruption tables

frame_domain_contingency_table

139	38
7	10

domain_disruption_descriptives

Domain_key	Domain_name	Quantitative_call_count	Fully_retained_count	Truncated_count	Fully_lost_count	Mean_retained_fraction_among_non_retained_calls
auto_disruption_pf00028	PF00028	194	17	1	176	0.003161151466236212

domain_retention tables

frame_domain_contingency_table

141	38
5	10

domain_retention_descriptives

Domain_key	Domain_name	Quantitative_call_count	Fully_retained_count	Truncated_count	Fully_lost_count	Mean_retained_fraction_among_non_retained_calls
kinase	Protein kinase domain	194	179	1	14	0.04988179669030733

exon_retention tables

Exon_retention

Exon	Retained_event_count	Total_event_count	Retained_fraction
12	180	194	0.9278350515463918

frequency tables

Partner_gene_counts

Partner_gene	Event_count
CCDC186	1
CCDC6	51
CLIP1	1
CTNNA1	1
CTNNA3	1
CUBN	1
DOCK1	1
EML4	1
ERC1	1
ERCC6	3
FBXL7	2
GEMIN5	1
GRIPAP1	1
KCNMA1	1
KIAA1217	1
KIF13A	1
KIF5B	87
LIN52	1
NCOA4	16
PDCD10	1
RASSF4	1
RBPMS	1
RELCH	1
RUFY1	2
RUFY2	1
RUFY3	2
SHROOM3	1
SLC4A4	1
SPECC1L	4
TFG	1
TIMM23B	3
TRIM24	1
UXS1	1

Per-event results (results.tsv)

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intron_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site_2_effect_on_frame
EVT-P-005 1781-T03-I M6-3	P-0051 781-T0 3-IM6		CCDC6-R ET	CCDC6	in-frame	threesome	43610377	11	712	True	retained	1.0	False	Protein kinase domain	PF00028		CCDC6 (NM_005436) - RET (NM_020975) fusion: c.303+14702:CCDC6_c.2136+193:RETinv Protein Fusion: in frame {CCDC6:RET}	NA
EVT-P-005 9857-T02-I M7-4	P-0059 857-T0 2-IM7		CCDC6-R ET	CCDC6	in-frame	threesome	43611615	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		CCDC6 (NM_005436) - RET (NM_020975) fusion: c.453+8003:CCDC6_c.2137-417:RETinv Protein Fusion: in frame {CCDC6:RET}	NA
EVT-P-006 5992-T01-I M7-5	P-0065 992-T0 1-IM7		CCDC6-R ET	CCDC6	in-frame	threesome	43611698	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		CCDC6 (NM_005436) - RET (NM_020975) fusion: c.303+12221:CCDC6_c.2137-334:RETinv Protein Fusion: in frame {CCDC6:RET}	NA
EVT-P-004 9461-T02-I M6-6	P-0049 461-T0 2-IM6		CCDC6-R ET	CCDC6	in-frame	threesome	43611705	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		CCDC6 (NM_005436) - RET (NM_020975) fusion: c.304-3450:CCDC6_c.2137-327:RETinv Protein Fusion: in frame {CCDC6:RET}	NA
EVT-P-003 6379-T01-I M6-7	P-0036 379-T0 1-IM6		CCDC6-R ET	CCDC6	in-frame	threesome	43611418	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		CCDC6 (NM_005436) - RET (NM_020975) fusion: c.303+26500:CCDC6_c.2137-614:RETinv Protein Fusion: in frame {CCDC6:RET}	NA
EVT-P-003 7412-T01-I M6-8	P-0037 412-T0 1-IM6		CCDC6-R ET	CCDC6	in-frame	threesome	43610839	11	712	True	retained	1.0	False	Protein kinase domain	PF00028		CCDC6 (NM_005436) - RET (NM_020975) fusion: c.304-11974:CCDC6_c.2136+655:RETinv Protein Fusion: in frame {CCDC6:RET}	NA
EVT-P-004 8611-T01-I M6-9	P-0048 611-T0 1-IM6		CCDC6-R ET	CCDC6	in-frame	threesome	43611237	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		CCDC6 (NM_005436) - RET (NM_020975) fusion: c.303+12938:CCDC6_c.2137-795:RETinv Protein Fusion: in frame {CCDC6:RET}	NA
EVT-P-006 4853-T01-I M7-10	P-0064 853-T0 1-IM7		CCDC6-R ET	CCDC6	out-of-frame	threesome	43609511	10	627	True	retained	1.0	False	Protein kinase domain	PF00028		CCDC6 (NM_005436) - RET (NM_020975) fusion: c.304-16584:CCDC6_c.1879+388:RETinv Protein Fusion: out of frame {CCDC6:RET}	NA
EVT-P-004 7144-T01-I M6-11	P-0047 144-T0 1-IM6		CCDC6-R ET	CCDC6	in-frame	threesome	43611888	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		CCDC6 (NM_005436) - RET (NM_020975) fusion: c.303+8847:CCDC6_c.2137-144:RETinv Protein Fusion: in frame {CCDC6:RET}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intron_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site_2_effect_on_frame
EVT-P-003 8257-T01-I M6-12	P-0038 257-T0 1-IM6		CCDC6-R ET	CCDC6	in-frame	threonine_prime	43610594	11	712	True	retained	1.0	False	Protein kinase domain	PF00028		CCDC6(NM_005436) - RET (NM_020975.4) Fusion (CCDC6 exon1 fused with RET exon12);c.304-2131:CCDC6_c.2136+410:RETinv Protein Fusion: in frame {CCDC6:RET}	NA
EVT-P-001 7365-T04-I M6-13	P-0017 365-T0 4-IM6		CCDC6-R ET	CCDC6	in-frame	threonine_prime	43611464	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		CCDC6 (NM_005436) - RET (NM_020975) fusion (CCDC6 exon 1 fused to RET exons 12-20): c.303+25501:CCDC6_c.2137-568:RETinv Protein Fusion: in frame {CCDC6:RET}	NA
EVT-P-003 9548-T01-I M6-14	P-0039 548-T0 1-IM6		CCDC6-R ET	CCDC6	in-frame	threonine_prime	43610843	11	712	True	retained	1.0	False	Protein kinase domain	PF00028		CCDC6 (NM_005436) - RET (NM_020975) fusion: c.304-9501:CCDC6_c.2136+659:RETinv Protein Fusion: in frame {CCDC6:RET}	NA
EVT-P-004 0111-T01-I M6-15	P-0040 111-T0 1-IM6		CCDC6-R ET	CCDC6	unknown	threonine_prime	43609967	11	640	False	retained	1.0	False	Protein kinase domain	PF00028		CCDC6 (NM_005436) - RET (NM_020975) fusion: c.304-13873:CCDC6_c.1919:RETinv Protein Fusion: mid-exon {CCDC6:RET}	NA
EVT-P-004 1192-T01-I M6-16	P-0041 192-T0 1-IM6		CCDC6-R ET	CCDC6	in-frame	threonine_prime	43610571	11	712	True	retained	1.0	False	Protein kinase domain	PF00028		CCDC6 (NM_005436) - RET (NM_020975) fusion: c.303+8632:CCDC6_c.2136+387:RETinv Protein Fusion: in frame {CCDC6:RET}	NA
EVT-P-002 2642-T04-I M6-17	P-0022 642-T0 4-IM6		CCDC6-R ET	CCDC6	in-frame	threonine_prime	43611768	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		CCDC6 (NM_005436) - RET (NM_020975) fusion: c.304-14213:CCDC6_c.2137-264:RETinv Protein Fusion: in frame {CCDC6:RET}	NA
EVT-P-002 2642-T03-I M6-18	P-0022 642-T0 3-IM6		CCDC6-R ET	CCDC6	in-frame	threonine_prime	43611768	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		CCDC6 (NM_005436) - RET (NM_020975) fusion: c.304-14213:CCDC6_c.2137-264:RETinv Protein Fusion: in frame {CCDC6:RET}	NA
EVT-P-005 1781-T01-I M6-19	P-0051 781-T0 1-IM6		CCDC6-R ET	CCDC6	in-frame	threonine_prime	43610377	11	712	True	retained	1.0	False	Protein kinase domain	PF00028		CCDC6 (NM_005436) - RET (NM_020975) fusion: c.303+14702:CCDC6_c.2136+193:RETinv Protein Fusion: in frame {CCDC6:RET}	NA
EVT-P-004 2417-T01-I M6-20	P-0042 417-T0 1-IM6		CCDC6-R ET	CCDC6	in-frame	threonine_prime	43610397	11	712	True	retained	1.0	False	Protein kinase domain	PF00028		CCDC6 (NM_005436) - RET (NM_020975) fusion: c.303+16388:CCDC6_c.2136+213:RETinv Protein Fusion: in frame {CCDC6:RET}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0004901-T01-I M5-21	P-0004901-T01-IM5		CCDC6-RET	CCDC6	in-frame	threonine_prime	43611729	12	713	True	retained	1.0	False	Protein_kinase_domain	PF00028		CCDC6 (NM_005436) - RET (NM_020975) reciprocal fusion (CCDC6 exon1 with RET exons 12-20) : :CCDC6_RETinv Protein fusion: in frame (CCDC6-RET)	NA
EVT-P-0022380-T03-I M6-22	P-0022380-T03-IM6		CCDC6-RET	CCDC6	in-frame	threonine_prime	43610659	11	712	True	retained	1.0	False	Protein_kinase_domain	PF00028		CCDC6 (NM_005436) - RET (NM_020975) fusion: c.303+21810:CCDC6_c.2136+475:RETinv Protein Fusion: in frame {CCDC6:RET}	NA
EVT-P-0068500-T01-I M7-23	P-0068500-T01-IM7		CCDC6-RET	CCDC6	in-frame	threonine_prime	43610322	11	712	True	retained	1.0	False	Protein_kinase_domain	PF00028		CCDC6 (NM_005436) - RET (NM_020975) fusion: c.303+11290:CCDC6_c.2136+138:RETinv Protein Fusion: in frame {CCDC6:RET}	NA
EVT-P-0057735-T01-I M6-24	P-0057735-T01-IM6		CCDC6-RET	CCDC6	in-frame	threonine_prime	43611887	12	713	True	retained	1.0	False	Protein_kinase_domain	PF00028		CCDC6 (NM_005436) - RET (NM_020975) fusion: c.303+125:CCDC6_c.2137-145:RETinv Protein Fusion: in frame {CCDC6:RET}	NA
EVT-P-0038053-T01-I M6-26	P-0038053-T01-IM6		CLIP1-RET	CLIP1	unknown	threonine_prime	43611121	12	713	True	retained	1.0	False	Protein_kinase_domain	PF00028		CLIP1 (NM_001247997) - RET (NM_020975) rearrangement: t(10;12)(q11.21;q24.31)(chr10:g.43611121::chr12:g.122825512) Protein Fusion: mid-exon {CLIP1:RET}	NA
EVT-P-0000252-T01-I M3-27	P-0000252-T01-IM3		CUBN-RET	CUBN	unknown	threonine_prime	43610494	11	712	True	retained	1.0	False	Protein_kinase_domain	PF00028		NA Antisense fusion	NA
EVT-P-0038256-T01-I M6-28	P-0038256-T01-IM6		ERC1-RET	ERC1	unknown	threonine_prime	43609964	11	639	False	retained	1.0	False	Protein_kinase_domain	PF00028		ERC1 (NM_178040) - RET (NM_020975) Rearrangement : t(10;12)(q11.1;p13.32)(chr10:g.43609964::chr12:g.1264106) Protein Fusion: mid-exon {ERC1:RET}	NA
EVT-P-0034781-T01-I M6-29	P-0034781-T01-IM6		GRIPAP1-RET	GRIPAP1	in-frame	threonine_prime	43611853	12	713	True	retained	1.0	False	Protein_kinase_domain	PF00028		GRIPAP1 (NM_020137) - RET (NM_020975) rearrangement: t(X;10)(p11.23q11.21)(chrX:g.48831008::chr10:g.43611853) Protein Fusion: in frame {GRIPAP1:RET}	NA
EVT-P-0026245-T01-I M6-30	P-0026245-T01-IM6		KIAA1217-RET	KIAA1217	unknown	threonine_prime	43606666	7	425	False	retained	1.0	False	Protein_kinase_domain	PF00028		KIAA1217 (NM_019590) - RET (NM_020975) fusion (KIAA1217 exons 1-13 fused to RET exons 7-20): c.2390:KIAA1217_c.1275:RETdel Protein Fusion: mid-exon {KIAA1217:RET}	NA

event_id	sampl_e_id	pa_tie_nt_id	fusio_n_name	part_ner_ge_ne	fra_me_st_atu_s	tar_get_ro_le	breakp_oint_ge_nomic	break_point_exo_n	breakpoint_protein_p_osition	is_intron_ic_break_point	dom_ain_statu_s	domain_r_etained_fr_action	domain_is_trun_cated	retain_ed_do_mains	lost_do_mains	disrupt_ed_do_mains	source_annotation_text	source_site_2_effect_on_frame
EVT-P-004 2153-T01-I M6-31	P-0042 153-T0 1-IM6		KIF1 3A-R ET	KIF 13A	in-fr ame	thr ee _pr im e	436105 10	11	712	True	retain ed	1.0	False	Protei n kinase domai n	PF0 002 8		KIF13A (NM_022113) - RET (NM_020975) fusion: t(6;10)(p22.3;q11.21)(chr6:g.17806558::chr10:g.43610510) Protein Fusion: in frame {KIF13A:RET}	NA
EVT-P-001 9535-T02-I M6-32	P-0019 535-T0 2-IM6		KIF5 B-RE T	KIF 5B	in-fr ame	thr ee _pr im e	436119 13	12	713	True	retain ed	1.0	False	Protei n kinase domai n	PF0 002 8		KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-16 fused to RET exons 12-20): c.1726-2113:KIF5B_c.2137-119:RETinv Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-003 9257-T03-I M7-33	P-0039 257-T0 3-IM7		KIF5 B-RE T	KIF 5B	unk now n	thr ee _pr im e	436100 60	11	671	False	retain ed	1.0	False	Protei n kinase domai n	PF0 002 8		KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+1132:KIF5B_c.2012:RETinv Protein Fusion: mid-exon {KIF5B:RET}	NA
EVT-P-000 9514-T01-I M5-34	P-0009 514-T0 1-IM5		KIF5 B-RE T	KIF 5B	in-fr ame	thr ee _pr im e	436118 11	12	713	True	retain ed	1.0	False	Protei n kinase domai n	PF0 002 8		Note: The KIF5B (NM_004521) - RET (NM_020975) reciprocal rearrangement event is an inversion which results in the fusion of KIF5B exons 1 to 15 and RET exons 12 to 20. The clinical trial research is currently ongoing with cabozantinib. PMID: 23533264 Protein fusion: in frame (KIF5B-RET)	NA
EVT-P-005 5183-T01-I M6-35	P-0055 183-T0 1-IM6		KIF5 B-RE T	KIF 5B	in-fr ame	thr ee _pr im e	436103 08	11	712	True	retain ed	1.0	False	Protei n kinase domai n	PF0 002 8		KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+1798:KIF5B_c.2136+124:RETinv Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-001 1035-T01-I M5-36	P-0011 035-T0 1-IM5		KIF5 B-RE T	KIF 5B	in-fr ame	thr ee _pr im e	436113 22	12	713	True	retain ed	1.0	False	Protei n kinase domai n	PF0 002 8		KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused with RET exons 12-20) : c.1726-2296:KIF5B_c.2137-710:RETinv Protein fusion: in frame (KIF5B-RET)	NA
EVT-P-001 2648-T01-I M5-37	P-0012 648-T0 1-IM5		KIF5 B-RE T	KIF 5B	in-fr ame	thr ee _pr im e	436116 97	12	713	True	retain ed	1.0	False	Protei n kinase domai n	PF0 002 8		KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+1863:KIF5B_c.2137-335:RETinv Protein fusion: in frame (KIF5B-RET)	NA
EVT-P-005 3175-T01-I M6-38	P-0053 175-T0 1-IM6		KIF5 B-RE T	KIF 5B	in-fr ame	thr ee _pr im e	436111 62	12	713	True	retain ed	1.0	False	Protei n kinase domai n	PF0 002 8		KIF5B (NM_004521) - RET (NM_020975) fusion: c.1726-282:KIF5B_c.2137-870:RETinv Protein Fusion: in frame {KIF5B:RET}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intron_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-005 2861-T02-I M6-39	P-0052 861-T0 2-IM6		KIF5 B-RE T	KIF 5B	in-frame	threesome	43610260	11	712	True	retained	1.0	False	Protein kinase domain	PF00028		KIF5B (NM_004521) - RET (NM_020975) fusion: c.1915-20:KIF5B_c.2136+76:RETinv Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-001 2825-T01-I M5-40	P-0013 825-T0 1-IM5		KIF5 B-RE T	KIF 5B	unknown	threesome	43611520	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-16 fused to RET exons 12-20): c.1782:KIF5B_c.2137-512:RETinv Protein fusion: mid-exon (KIF5B-RET)	NA
EVT-P-005 2826-T01-I M6-41	P-0052 826-T0 1-IM6		KIF5 B-RE T	KIF 5B	in-frame	threesome	43611182	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		KIF5B (NM_004521) - RET (NM_020975) fusion: c.1726-2358:KIF5B_c.2137-850:RETinv Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-001 4808-T01-I M6-42	P-0014 808-T0 1-IM6		KIF5 B-RE T	KIF 5B	in-frame	threesome	43611055	11	712	True	retained	1.0	False	Protein kinase domain	PF00028		KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused to RET exons 12-20): c.1725+1765:KIF5B_c.2136+871:RETinv Protein fusion: in frame (KIF5B-RET)	NA
EVT-P-003 4357-T01-I M6-43	P-0034 357-T0 1-IM6		KIF5 B-RE T	KIF 5B	in-frame	threesome	43611187	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+1863:KIF5B_c.2137-845:RETinv Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-001 4808-T02-I M6-44	P-0014 808-T0 2-IM6		KIF5 B-RE T	KIF 5B	in-frame	threesome	43611056	11	712	True	retained	1.0	False	Protein kinase domain	PF00028		KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused to RET exons 12-20): c.1725+1765:KIF5B_c.2136+871:RETinv Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-003 3700-T02-I M6-45	P-0033 700-T0 2-IM6		KIF5 B-RE T	KIF 5B	in-frame	threesome	43611670	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+1454:KIF5B_c.2137-362:RET Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-001 4808-T03-I M6-46	P-0014 808-T0 3-IM6		KIF5 B-RE T	KIF 5B	in-frame	threesome	43611056	11	712	True	retained	1.0	False	Protein kinase domain	PF00028		KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+1764:KIF5B_c.2136+872:RETinv Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-000 8289-T01-I M5-47	P-0008 289-T0 1-IM5		KIF5 B-RE T	KIF 5B	in-frame	threesome	43611748	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused to RET exons 12-20) : c.1726-824:KIF5B_c.2137-284:RETinv Protein fusion: in frame (KIF5B-RET)	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intron_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site_2_effect_on_frame
EVT-P-000 0252-T01-I M3-48	P-0000 252-T0 1-IM3		KIF5 B-RET	KIF5B	in-frame	threesome_prime	43610597	11	712	True	retained	1.0	False	Protein kinase domain	PF00028		RET (NM_020975) Inversion (c.2136+413_KIF5B:c.1726-153inv) Protein fusion: in frame (KIF5B-RET)	NA
EVT-P-000 8101-T01-I M5-49	P-0008 101-T0 1-IM5		KIF5 B-RET	KIF5B	in-frame	threesome_prime	43610455	11	712	True	retained	1.0	False	Protein kinase domain	PF00028		KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-18 fused with RET exons 12-20) : c.2095-38:KIF5B_c.2136+271:RETinv Protein fusion: in frame (KIF5B-RET)	NA
EVT-P-004 8968-T01-I M6-50	P-0048 968-T0 1-IM6		KIF5 B-RET	KIF5B	in-frame	threesome_prime	43611809	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		KIF5B (NM_004521) - RET (NM_020975) fusion: c.1914+211:KIF5B_c.2137-223:RETinv Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-004 8829-T01-I M6-51	P-0048 829-T0 1-IM6		KIF5 B-RET	KIF5B	in-frame	threesome_prime	43610616	11	712	True	retained	1.0	False	Protein kinase domain	PF00028		KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+948:KIF5B_c.2136+432:RETinv Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-006 5876-T01-I M7-52	P-0065 876-T0 1-IM7		KIF5 B-RET	KIF5B	in-frame	threesome_prime	43611299	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		KIF5B (NM_004521) - RET (NM_020975) fusion: c.1726-1902:KIF5B_c.2137-733:RETinv Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-001 6221-T01-I M6-53	P-0016 221-T0 1-IM6		KIF5 B-RET	KIF5B	in-frame	threesome_prime	43611268	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused in-frame to RET exons 12-20) : c.1725+275:KIF5B_c.2137-764:RETinv Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-003 7367-T01-I M6-54	P-0037 367-T0 1-IM6		KIF5 B-RET	KIF5B	in-frame	threesome_prime	43611874	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+1038:KIF5B_c.2137-158:RETinv Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-005 7072-T02-I M6-55	P-0057 072-T0 2-IM6		KIF5 B-RET	KIF5B	in-frame	threesome_prime	43611106	11	712	True	retained	1.0	False	Protein kinase domain	PF00028		KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+2508:KIF5B_c.2136+922:RETinv Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-001 7214-T01-I M6-56	P-0017 214-T0 1-IM6		KIF5 B-RET	KIF5B	in-frame	threesome_prime	43610647	11	712	True	retained	1.0	False	Protein kinase domain	PF00028		KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-23 fused in-frame with RET exons 12-20): c.2545-27:KIF5B_c.2136+463:RETinv Protein Fusion: in frame {KIF5B:RET}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intron_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site_2_effect_on_frame
EVT-P-001 7214-T02-I M6-57	P-0017 214-T0 2-IM6		KIF5 B-RET	KIF5B	in-frame	threonine_prime	43610647	11	712	True	retained	1.0	False	Protein kinase domain	PF00028		KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-23 fused to RET exons 12-20): c.2545-27:KIF5B_c.2136+463:RETinv Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-003 2753-T01-I M6-58	P-0032 753-T0 1-IM6		KIF5 B-RET	KIF5B	in-frame	threonine_prime	43610925	11	712	True	retained	1.0	False	Protein kinase domain	PF00028		KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+1323:KIF5B_c.2136+741:RETinv Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-005 7072-T03-I M7-59	P-0057 072-T0 3-IM7		KIF5 B-RET	KIF5B	in-frame	threonine_prime	43611106	11	712	True	retained	1.0	False	Protein kinase domain	PF00028		KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+2508:KIF5B_c.2136+922:RETinv Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-003 2686-T01-I M6-60	P-0032 686-T0 1-IM6		KIF5 B-RET	KIF5B	in-frame	threonine_prime	43611982	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+2299:KIF5B_c.2137-50:RETinv Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-001 7523-T02-I M6-61	P-0017 523-T0 2-IM6		KIF5 B-RET	KIF5B	in-frame	threonine_prime	43611192	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused to RET exons 12-20): c.1726-2362:KIF5B_c.2137-840:RETinv Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-004 6827-T02-I M6-62	P-0046 827-T0 2-IM6		KIF5 B-RET	KIF5B	in-frame	threonine_prime	43610889	11	712	True	retained	1.0	False	Protein kinase domain	PF00028		KIF5B (NM_004521) - RET (NM_020975) fusion: c.1726-1305:KIF5B_c.2136+705:RETinv Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-003 2407-T01-I M6-63	P-0032 407-T0 1-IM6		KIF5 B-RET	KIF5B	in-frame	threonine_prime	43611730	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		KIF5B (NM_004521) - RET (NM_020975) fusion: c.1914+230:KIF5B_c.2137-302:RETinv Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-001 8069-T01-I M6-64	P-0018 069-T0 1-IM6		KIF5 B-RET	KIF5B	out-of-frame	five_prime	43607832	8	550	True	lost	0.0	False	Protein kinase domain	PF00028		RET (NM_020975) - KIF5B (NM_004521) rearrangement: c.1648+160:RET_c.1915-72:KIF5Binv Protein Fusion: out of frame {RET:KIF5B}	NA
EVT-P-000 5573-T01-I M5-65	P-0005 573-T0 1-IM5		KIF5 B-RET	KIF5B	in-frame	threonine_prime	43610418	11	712	True	retained	1.0	False	Protein kinase domain	PF00028		KIF5B (NM_004521) - RET (NM_020975) Fusion (KIF5B exon15 fused with RET exon12): c.1726-860:KIF5B_c.2136+234:RETinv Protein fusion: in frame (KIF5B-RET)	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intron_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0004995-T01-I M5-66	P-0004995-T01-IM5		KIF5B-RET	KIF5B	in-frame	threesome	43610660	11	712	True	retained	1.0	False	Protein kinase domain	PF00028		KIF5B(NM_004521) -RET (NM_020975) Fusion (KIF5B exon 15 fused to RET exon 12) : c.1726-2366:KIF5B_c.2136+476:RETinv Protein fusion: in frame (KIF5B-RET)	NA
EVT-P-0036092-T01-I M6-67	P-0036092-T01-IM6		KIF5B-RET	KIF5B	in-frame	threesome	43610413	11	712	True	retained	1.0	False	Protein kinase domain	PF00028		KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused to RET exons 12-20): c.1725+1138:KIF5B_c.2136+229:RETinv Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0046726-T01-I M6-68	P-0046726-T01-IM6		KIF5B-RET	KIF5B	in-frame	threesome	43610987	11	712	True	retained	1.0	False	Protein kinase domain	PF00028		KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+768:KIF5B_c.2136+803:RETinv Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0032227-T01-I M6-69	P-0032227-T01-IM6		KIF5B-RET	KIF5B	in-frame	threesome	43611125	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+2000:KIF5B_c.2137-907:RETinv Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0020273-T02-I M6-70	P-0020273-T02-IM6		KIF5B-RET	KIF5B	in-frame	threesome	43611819	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-16 fused to RET exons 12-20): c.1914+131:KIF5B_c.2137-213:RETinv Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0043007-T02-I M6-71	P-0043007-T02-IM6		KIF5B-RET	KIF5B	in-frame	threesome	43610813	11	712	True	retained	1.0	False	Protein kinase domain	PF00028		KIF5B (NM_004521) - RET (NM_020975) fusion: c.2544+168:KIF5B_c.2136+629:RETinv Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0021150-T01-I M6-72	P-0021150-T01-IM6		KIF5B-RET	KIF5B	in-frame	threesome	43611591	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused to RET exons 12-20): c.1725+312:KIF5B_c.2137-441:RETinv Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0021578-T01-I M6-73	P-0021578-T01-IM6		KIF5B-RET	KIF5B	in-frame	threesome	43610809	11	712	True	retained	1.0	False	Protein kinase domain	PF00028		KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused to RET exons 12-20): c.1725+1231:KIF5B_c.2136+624:RETinv Protein Fusion: in frame {KIF5B:RET}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intron_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site_2_effect_on_frame
EVT-P-002 1879-T01-I M6-74	P-0021 879-T0 1-IM6		KIF5 B-RET	KIF5B	in-frame	threonine_prime	43610699	11	712	True	retained	1.0	False	Protein kinase domain	PF00028		KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B1 exons 1-15 fused with RET exons 12-20 including the kinase domain.): c.1726-882:KIF5B_c.2136+515_RETinv Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-002 1937-T01-I M6-75	P-0021 937-T0 1-IM6		KIF5 B-RET	KIF5B	in-frame	threonine_prime	43611555	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused with RET exons 12-20) : c.1725+886:KIF5B_c.2137-477_RETinv Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-002 1937-T03-I M6-76	P-0021 937-T0 3-IM6		KIF5 B-RET	KIF5B	in-frame	threonine_prime	43611555	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+886:KIF5B_c.2137-477_RETinv Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-004 3007-T01-I M6-77	P-0043 007-T0 1-IM6		KIF5 B-RET	KIF5B	in-frame	threonine_prime	43610813	11	712	True	retained	1.0	False	Protein kinase domain	PF00028		KIF5B (NM_004521) - RET (NM_020975) fusion: c.2544+168:KIF5B_c.2136+629_RETinv Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-004 2608-T01-I M6-78	P-0042 608-T0 1-IM6		KIF5 B-RET	KIF5B	in-frame	threonine_prime	43611802	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		KIF5B (NM_004521) - RET (NM_020975) fusion: c.1726-1670:KIF5B_c.2137-230_RETinv Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-004 2580-T01-I M6-79	P-0042 580-T0 1-IM6		KIF5 B-RET	KIF5B	in-frame	threonine_prime	43610944	11	712	True	retained	1.0	False	Protein kinase domain	PF00028		KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+1619:KIF5B_c.2136+760_RETinv Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-005 9913-T01-I M7-80	P-0059 913-T0 1-IM7		KIF5 B-RET	KIF5B	unknown	threonine_prime	43610108	11	687	False	retained	1.0	False	Protein kinase domain	PF00028		KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+2441:KIF5B_c.2060_RETinv Protein Fusion: mid-exon {KIF5B:RET}	NA
EVT-P-000 1010-T02-I M5-81	P-0001 010-T0 2-IM5		KIF5 B-RET	KIF5B	in-frame	threonine_prime	43610863	11	712	True	retained	1.0	False	Protein kinase domain	PF00028		KIF5B (NM_004521) - RET (NM_020975) fusion: c.1726-780:KIF5B_c.2136+679_RETinv Protein fusion: in frame (KIF5B-RET)	NA
EVT-P-006 0748-T02-I M7-82	P-0060 748-T0 2-IM7		KIF5 B-RET	KIF5B	in-frame	threonine_prime	43611527	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		KIF5B (NM_004521) - RET (NM_020975) fusion: c.1726-2304:KIF5B_c.2137-505_RETinv Protein Fusion: in frame {KIF5B:RET}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site_2_effect_on_frame
EVT-P-004 2104-T01-I M6-83	P-0042 104-T0 1-IM6		KIF5 B-RET	KIF5B	in-frame	three_prime	43610505	11	712	True	retained	1.0	False	Protein kinase domain	PF00028		KIF5B (NM_004521) - RET (NM_020975) fusion: c.1726-1056:KIF5B_c.2136+321:RETinv Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-000 3125-T01-I M5-84	P-0003 125-T0 1-IM5		KIF5 B-RET	KIF5B	in-frame	three_prime	43611584	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		NA Protein fusion: in frame (KIF5B-RET)	NA
EVT-P-000 3125-T01-I M5-85	P-0003 125-T0 1-IM5		KIF5 B-RET	KIF5B	in-frame	five_prime	43611581	12	713	True	lost	0.0	False	PF00028	Protein kinase domain		NA Protein fusion: in frame (RET-KIF5B)	NA
EVT-P-000 2873-T01-I M3-86	P-0002 873-T0 1-IM3		KIF5 B-RET	KIF5B	in-frame	three_prime	43611244	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		NA Protein fusion: in frame (KIF5B-RET)	NA
EVT-P-002 2866-T01-I M6-87	P-0022 866-T0 1-IM6		KIF5 B-RET	KIF5B	in-frame	three_prime	43611871	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		KIF5B (NM_004521) - RET (NM_020975) Fusion (KIF5B exons 1-15 fused to RET exons 12-20): c.1726-767:KIF5B_c.2137-161:RETinv Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-002 4225-T01-I M6-88	P-0024 694-T0 1-IM6		KIF5 B-RET	KIF5B	unknown	three_prime	43609110	10	622	False	retained	1.0	False	Protein kinase domain	PF00028		KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-24 fused to RET exons 10-20): c.2762-430:KIF5B_c.1866:RETinv Protein Fusion: mid-exon {KIF5B:RET}	NA
EVT-P-000 2694-T02-I M5-89	P-0002 694-T0 2-IM5		KIF5 B-RET	KIF5B	in-frame	three_prime	43611682	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		NA Protein fusion: in frame (KIF5B-RET)	NA
EVT-P-002 4843-T01-I M6-90	P-0024 843-T0 1-IM6		KIF5 B-RET	KIF5B	in-frame	three_prime	43611733	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 with RET exons 12-20): c.1726-1339:KIF5B_c.2137-299:RETinv Protein Fusion: in frame {KIF5B:RET}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intron_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site_2_effect_on_frame
EVT-P-002 5100-T01-I M6-91	P-0025 100-T0 1-IM6		KIF5 B-RET	KIF5B	in-frame	threep _prime	436115 11	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused to RET exons 12-20): c.1726-2273:KIF5B_c.2137-521:RETinv Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-002 5100-T02-I M6-92	P-0025 100-T0 2-IM6		KIF5 B-RET	KIF5B	in-frame	threep _prime	436115 11	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		KIF5B (NM_004521) and RET (NM_020975) Fusion (KIF5B exon 16 fused to RET exon 12): c.1726-2273:KIF5B_c.2137-788:RETinv Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-003 9941-T01-I M6-93	P-0039 941-T0 1-IM6		KIF5 B-RET	KIF5B	in-frame	threep _prime	436113 39	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+2040:KIF5B_c.2137-693:RETinv Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-006 0748-T03-I M7-94	P-0060 748-T0 3-IM7		KIF5 B-RET	KIF5B	in-frame	threep _prime	436115 27	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		KIF5B (NM_004521) - RET (NM_020975) fusion: c.1726-2304:KIF5B_c.2137-505:RETinv Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-003 9474-T02-I M6-95	P-0039 474-T0 2-IM6		KIF5 B-RET	KIF5B	in-frame	threep _prime	436103 59	11	712	True	retained	1.0	False	Protein kinase domain	PF00028		KIF5B (NM_004521) - RET (NM_020975) fusion: c.1915-203:KIF5B_c.2136+175:RETinv Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-003 9474-T01-I M6-96	P-0039 474-T0 1-IM6		KIF5 B-RET	KIF5B	in-frame	threep _prime	436103 59	11	712	True	retained	1.0	False	Protein kinase domain	PF00028		KIF5B (NM_004521) - RET (NM_020975) fusion: c.1915-203:KIF5B_c.2136+175:RETinv Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-005 5183-T02-I M6-97	P-0055 183-T0 2-IM6		KIF5 B-RET	KIF5B	in-frame	threep _prime	436103 08	11	712	True	retained	1.0	False	Protein kinase domain	PF00028		KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+1798:KIF5B_c.2136+124:RETinv Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-002 5838-T01-I M6-98	P-0025 838-T0 1-IM6		KIF5 B-RET	KIF5B	in-frame	threep _prime	436109 72	11	712	True	retained	1.0	False	Protein kinase domain	PF00028		KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused with RET exons 12-20.): c.1726-601:KIF5B_c.2136+788:RETinv Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-002 5928-T01-I M6-99	P-0025 928-T0 1-IM6		KIF5 B-RET	KIF5B	in-frame	threep _prime	436119 10	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused to RET exons 12-20): c.1725+483:KIF5B_c.2137-122C>T:RETinv Protein Fusion: in frame {KIF5B:RET}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intron_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site_2_effect_on_frame
EVT-P-002 5928-T02-I M6-100	P-0025 928-T0 2-IM6		KIF5 B-RET	KIF5B	in-frame	three_prime	43611910	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+483:KIF5B_c.2137-122:RETinv Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-003 6637-T01-I M6-101	P-0036 637-T0 1-IM6		KIF5 B-RET	KIF5B	in-frame	three_prime	43611965	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused to RET exons 12-20): c.1726-1731:KIF5B_c.2137-67:RETinv Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-003 9257-T01-I M6-102	P-0039 257-T0 1-IM6		KIF5 B-RET	KIF5B	unknown	three_prime	43610060	11	671	False	retained	1.0	False	Protein kinase domain	PF00028		KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+1132:KIF5B_c.2012:RETinv Protein Fusion: mid-exon {KIF5B:RET}	NA
EVT-P-002 6693-T01-I M6-103	P-0026 693-T0 1-IM6		KIF5 B-RET	KIF5B	in-frame	three_prime	43610511	11	712	True	retained	1.0	False	Protein kinase domain	PF00028		KIF5B (NM_004521) - RET (NM_020975) fusion: c.2544+156:KIF5_c.2136+327:RETinv. Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-002 6777-T01-I M6-104	P-0026 777-T0 1-IM6		KIF5 B-RET	KIF5B	in-frame	three_prime	43610717	11	712	True	retained	1.0	False	Protein kinase domain	PF00028		KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-16 fused to RET exons 12-20): c.1726-1554:KIF5B_c.2136+533:RETinv Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-002 6777-T02-I M6-105	P-0026 777-T0 2-IM6		KIF5 B-RET	KIF5B	in-frame	three_prime	43610717	11	712	True	retained	1.0	False	Protein kinase domain	PF00028		KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-16 fused to RET exons 12-20): c.1726-1554:KIF5B_c.2136+533:RETinv Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-003 8961-T01-I M6-106	P-0038 961-T0 1-IM6		KIF5 B-RET	KIF5B	in-frame	three_prime	43610775	11	712	True	retained	1.0	False	Protein kinase domain	PF00028		KIF5B (NM_004521) - RET (NM_020975) fusion: c.1726-293:KIF5B_c.2136+591:RETinv Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-002 7723-T01-I M6-107	P-0027 723-T0 1-IM6		KIF5 B-RET	KIF5B	unknown	three_prime	43610171	11	708	False	retained	1.0	False	Protein kinase domain	PF00028		KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused to RET exons 11-20): c.1725+966:KIF5B_c.2123:RETinv Protein Fusion: mid-exon {KIF5B:RET}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intron_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site_2_effect_on_frame
EVT-P-003 6737-T02-I M6-108	P-0036 737-T0 2-IM6		KIF5 B-RET	KIF5B	in-frame	threonine_prime	43612025	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused to RET exons 12-20): c.1725+1487:KIF5B_c.2137-7:RETinv Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-006 1331-T01-I M7-109	P-0061 331-T0 1-IM7		KIF5 B-RET	KIF5B	in-frame	threonine_prime	43610884	11	712	True	retained	1.0	False	Protein kinase domain	PF00028		KIF5B (NM_004521) - RET (NM_020975) fusion: c.1915-286:KIF5B_c.2136+700:RETinv Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-002 7991-T01-I M6-110	P-0027 991-T0 1-IM6		KIF5 B-RET	KIF5B	in-frame	threonine_prime	43608201	9	550	True	retained	1.0	False	Protein kinase domain	PF00028		KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-24 fused to RET exons 9-20): c.2762-248:KIF5B_c.1649-100:RETinv Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-006 2948-T03-I M7-111	P-0062 948-T0 3-IM7		KIF5 B-RET	KIF5B	in-frame	threonine_prime	43610211	11	712	True	retained	1.0	False	Protein kinase domain	PF00028		KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+1625:KIF5B_c.2136+27:RETinv Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-006 4951-T01-I M7-112	P-0064 951-T0 1-IM7		KIF5 B-RET	KIF5B	out-of-frame	threonine_prime	43611130	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		KIF5B (NM_004521) - RET (NM_020975) fusion: c.2204+2:KIF5B_c.2137-902:RETinv Protein Fusion: out of frame {KIF5B:RET}	NA
EVT-P-003 0453-T01-I M6-113	P-0030 453-T0 1-IM6		KIF5 B-RET	KIF5B	in-frame	threonine_prime	43610911	11	712	True	retained	1.0	False	Protein kinase domain	PF00028		KIF5B (NM_004521) - RET (NM_020975) fusion: c.1726-2396:KIF5B_c.2136+727:RETinv Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-003 7494-T01-I M6-114	P-0037 494-T0 1-IM6		KIF5 B-RET	KIF5B	in-frame	threonine_prime	43611755	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		KIF5B (NM_004521) - RET (NM_020975) Fusion (KIF5B exon15 fused with RET exon12): c.1725+985:KIF5B_c.2137-277:RETinv Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-000 0208-T02-I M5-115	P-0000 208-T0 2-IM5		KIF5 B-RET	KIF5B	in-frame	threonine_prime	43611647	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused with RET exons 12- 20): c.1726-1971:KIF5B_c.2137-385:RETinv Protein fusion: in frame (KIF5B-RET)	NA
EVT-P-000 0208-T01-I M3-116	P-0000 208-T0 1-IM3		KIF5 B-RET	KIF5B	in-frame	threonine_prime	43611647	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		RET (NM_020975) Inversion (c.2137-385_KIF5B:c.1726-1971inv) Protein fusion: in frame (KIF5B-RET)	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intron_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site_2_effect_on_frame
EVT-P-001 6221-T02-I M6-117	P-0016 221-T0 2-IM6		KIF5B-RET	KIF5B	in-frame	threonine_prime	43611268	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 with RET exons 12-20 including the kinase domain): c.1725+275:KIF5B_c.2137-764:RETinv Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-001 9663-T01-I M6-118	P-0019 663-T0 1-IM6		LIN52-RET	LIN52	out-of-frame	five_prime	43608613	9	587	True	lost	0.0	False	PF00028	Protein kinase domain		RET (NM_020975) - LIN52 (NM_001024674) Rearrangement : t(10;14)(p32.1;q12)(chr10:g.43608613:chr14:g.74562750) Protein Fusion: out of frame {RET:LIN52}	NA
EVT-P-006 5826-T01-I M7-120	P-0065 826-T0 1-IM7		NCOA4-RET	NCOA4	out-of-frame	threonine_prime	43610857	11	712	True	retained	1.0	False	Protein kinase domain	PF00028		NCOA4 (NM_001145260) - RET (NM_020975) fusion: c.2136+673:RET_c.*23+937NCOA4dup Protein Fusion: out of frame {TIMM23B:RET}	NA
EVT-P-003 7961-T01-I M6-121	P-0037 961-T0 1-IM6		NCOA4-RET	NCOA4	out-of-frame	threonine_prime	43611448	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		NCOA4 (NM_001145260) - RET (NM_020975) rearrangement: c.763-360:NCOA4_c.2137-584:RETdup Protein Fusion: out of frame {TIMM23B:RET}	NA
EVT-P-003 8315-T01-I M6-122	P-0038 315-T0 1-IM6		NCOA4-RET	NCOA4	out-of-frame	threonine_prime	43611981	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		NCOA4 (NM_001145260) - RET (NM_020975) fusion: c.763-415:NCOA4_c.2137-51:RETdup Protein Fusion: out of frame {TIMM23B:RET}	NA
EVT-P-006 0765-T02-I M7-123	P-0060 765-T0 2-IM7		NCOA4-RET	NCOA4	out-of-frame	threonine_prime	43611844	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		NCOA4 (NM_001145260) - RET (NM_020975) fusion: c.2137-188:RET_c.763-448NCOA4dup Protein Fusion: out of frame {TIMM23B:RET}	NA
EVT-P-004 3077-T01-I M6-124	P-0043 077-T0 1-IM6		NCOA4-RET	NCOA4	out-of-frame	threonine_prime	43611580	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		NCOA4 (NM_001145260) - RET (NM_020975) fusion: c.762+608:NCOA4_c.2137-452:RETdup Protein Fusion: out of frame {TIMM23B:RET}	NA
EVT-P-000 5748-T01-I M5-125	P-0005 748-T0 1-IM5		NCOA4-RET	NCOA4	in-frame	threonine_prime	43611325	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		NCOA4 (NM_001145260) - RET (NM_020975) fusion : c.763-260:NCOA4_c.2137-707:RETdup Protein fusion: in frame (NCOA4-RET)	NA
EVT-P-004 7991-T02-I M6-126	P-0047 991-T0 2-IM6		NCOA4-RET	NCOA4	out-of-frame	threonine_prime	43611538	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		NCOA4 (NM_001145260) - RET (NM_020975) fusion: c.762+537:NCOA4_c.2137-494:RETdup Protein Fusion: out of frame {TIMM23B:RET}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-004 8805-T01-I M6-127	P-0048 805-T0 1-IM6		NCOA4-R ET	NCOA4	unknown	threesome	43611225	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		NCOA4 (NM_001145260) - RET (NM_020975) fusion: c.851:NCOA4_c.2137-807:RETdup Protein Fusion: mid-exon {NCOA4:RET}	NA
EVT-P-004 9600-T01-I M6-128	P-0049 600-T0 1-IM6		NCOA4-R ET	NCOA4	unknown	threesome	43611613	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		NCOA4 (NM_001145260) - RET (NM_020975) fusion: c.1889:NCOA4_c.2137-419:RETdup Protein Fusion: mid-exon {NCOA4:RET}	NA
EVT-P-005 6870-T01-I M6-129	P-0056 870-T0 1-IM6		NCOA4-R ET	NCOA4	out-of-frame	threesome	43610337	11	712	True	retained	1.0	False	Protein kinase domain	PF00028		NCOA4 (NM_001145260) - RET (NM_020975) fusion: c.618+132:NCOA4_c.2136+153:RETdup Protein Fusion: out of frame {TIMM23B:RET}	NA
EVT-P-005 2409-T01-I M6-130	P-0052 409-T0 1-IM6		NCOA4-R ET	NCOA4	out-of-frame	threesome	43611691	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		NCOA4 (NM_001145260) - RET (NM_020975) rearrangement: c.618+214:NCOA4_c.2137-341:RETdup Protein Fusion: out of frame {TIMM23B:RET}	NA
EVT-P-000 4094-T02-I M6-131	P-0004 094-T0 2-IM6		RELCH-R ET	RELCH	in-frame	threesome	43610279	11	712	True	retained	1.0	False	Protein kinase domain	PF00028		KIAA1468 (NM_020854) - RET (NM_020975) fusion (KIAA1468 exons 1-10 fused with RET exons 12-20): t(10;18)(q11.21;q21.33)(chr10:g.43610279::chr18:g.59909399) Protein Fusion: in frame {KIAA1468:RET}	NA
EVT-P-001 5588-T01-I M6-132	P-0015 588-T0 1-IM6		RET-CCD C186	CCDC186	in-frame	threesome	43610308	11	712	True	retained	1.0	False	Protein kinase domain	PF00028		C10orf118 (NM_018017) - RET (NM_020975) Rearrangement : c.1326+142:C10orf118_c.2136+124:RETinv Protein fusion: in frame (C10orf118-RET)	NA
EVT-P-002 0974-T01-I M6-133	P-0020 974-T0 1-IM6		RET-CCD C6	CCDC6	in-frame	threesome	43611184	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		CCDC6 (NM_005436) - RET (NM_020975) fusion (CCDC6 exon 1 fused to RET exon 12): c.303+2543:CCDC6_c.2137-848:RETinv Protein Fusion: in frame {CCDC6:RET}	NA
EVT-P-000 1745-T02-I M5-134	P-0001 745-T0 2-IM5		RET-CCD C6	CCDC6	in-frame	threesome	43611245	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		CCDC6 (NM_005436) - (RET (NM_020975) rearrangement: c.304-9458:CCDC6_c.2137-787:RETinv Protein fusion: in frame (CCDC6-RET)	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site_2_effect_on_frame
EVT-P-001 7365-T02-I M6-135	P-0017 365-T0 2-IM6		RET- CCD C6	CC DC6	in-frame	three_prime	436114 64	12	713	True	retained	1.0	False	Protein kinase domain	PF0 002 8		CCDC6 (NM_005436) - RET (NM_020975) fusion (CCDC6 exon 1 fused to RET exons 12-20): c.303+25501:CCDC6_c.2137-568:RETinv Protein Fusion: in frame {CCDC6:RET}	NA
EVT-P-000 1898-T01-I M3-136	P-0001 898-T0 1-IM3		RET- CCD C6	CC DC6	in-frame	three_prime	436116 99	12	713	True	retained	1.0	False	Protein kinase domain	PF0 002 8		CCDC6 (NM_005436) - RET (NM_020975) inversion: c.304-12339_c.2137-333inv5 Protein fusion: in frame (CCDC6-RET)	NA
EVT-P-002 9602-T01-I M6-137	P-0029 602-T0 1-IM6		RET- CCD C6	CC DC6	in-frame	three_prime	436111 60	12	713	True	retained	1.0	False	Protein kinase domain	PF0 002 8		CCDC6 (NM_005436) - RET (NM_020975) fusion: c.304-5768:CCDC6_c.2137-872:RETinv Protein Fusion: in frame {CCDC6:RET}	NA
EVT-P-000 4916-T01-I M5-138	P-0004 916-T0 1-IM5		RET- CCD C6	CC DC6	in-frame	three_prime	436103 53	11	712	True	retained	1.0	False	Protein kinase domain	PF0 002 8		CCDC6(NM_005436)-RET(NM_020975) Fusion(CCDC6 exon 2 fused to RET exon11) : c.304-1153:CCDC6_c.2136+169:RETinv Protein fusion: in frame (CCDC6-RET)	NA
EVT-P-000 0425-T01-I M3-139	P-0000 425-T0 1-IM3		RET- CCD C6	CC DC6	in-frame	five_prime	436104 50	11	712	True	lost	0.0	False	PF00 28	Protein kinase domain		RET (NM_020630) - CCDC6 (NM_005436) Inversion (c.2136+266_c.303+19630inv) Protein fusion: in frame (RET-CCDC6)	NA
EVT-P-003 1330-T01-I M6-140	P-0031 330-T0 1-IM6		RET- CCD C6	CC DC6	in-frame	three_prime	436103 53	11	712	True	retained	1.0	False	Protein kinase domain	PF0 002 8		CCDC6 (NM_005436) - RET (NM_020975) fusion (CCDC6 exon 1 fused to RET exons 12-20): c.303+3703:CCDC6_c.2136+169:RETinv Protein Fusion: in frame {CCDC6:RET}	NA
EVT-P-003 0990-T03-I M6-141	P-0030 990-T0 3-IM6		RET- CCD C6	CC DC6	in-frame	three_prime	436118 81	12	713	True	retained	1.0	False	Protein kinase domain	PF0 002 8		CCDC6 (NM_005436) - RET (NM_020975) fusion (CCDC6 exon 1 fused to RET exon 12-20): c.303+10940:CCDC6_c.2137-151:RETinv Protein Fusion: in frame {CCDC6:RET}	NA
EVT-P-000 5168-T01-I M5-142	P-0005 168-T0 1-IM5		RET- CCD C6	CC DC6	out-of-frame	three_prime	436087 97	10	587	True	retained	1.0	False	Protein kinase domain	PF0 002 8		CCDC6 (NM_005436) - RET (NM_020975) Fusion (CCDC6 Exon1 fused to RET Exon10) : c.304-11493:CCDC6_c.1760-207:RETinv Protein fusion: out of frame (CCDC6-RET)	NA

event_id	sampl_e_id	pa_tie_n_t_i_d	fusio_n_na_me	part_ner_ge_ne	fra_me_st_atu_s	tar_get_ro_le	breakp_o_int_ge_nomic	break_p_o_int_exo_n	breakpoint_protein_p_osition	is_intron_ic_break_p_o_int	dom_ain_statu_s	domain_r_etained_fr_action	domain_is_tru_n_cated	retain_ed_do_mains	lost_do_mains	disrupt_ed_do_mains	source_annotation_text	source_site_2_effect_on_frame
EVT-P-002 2380-T01-I M6-143	P-0022 380-T0 1-IM6		RET- CCD C6	CC DC6	in-fr ame	thr_ee _pr im_e	436106 59	11	712	True	retain ed	1.0	False	Protei n kinase domai n	PF0 002 8		CCDC6 (NM_005436) - RET (NM_020975) fusion (CCDC6 exon 1 fused to RET exons 12-20): c.303+21810:CCDC6_c.2136+475:RETinv Protein Fusion: in frame {CCDC6:RET}	NA
EVT-P-002 2380-T02-I M6-144	P-0022 380-T0 2-IM6		RET- CCD C6	CC DC6	in-fr ame	thr_ee _pr im_e	436106 59	11	712	True	retain ed	1.0	False	Protei n kinase domai n	PF0 002 8		CCDC6 (NM_005436) - RET(NM_020975) fusion (CCDC6 exon 1 fused to RET exons 12-20): c.303+21810:CCDC6_c.2136+475:c.2136+475inv Protein Fusion: in frame {CCDC6:RET}	NA
EVT-P-002 9315-T02-I M6-145	P-0029 315-T0 2-IM6		RET- CCD C6	CC DC6	in-fr ame	thr_ee _pr im_e	436106 31	11	712	True	retain ed	1.0	False	Protei n kinase domai n	PF0 002 8		CCDC6 (NM_005436) - RET (NM_020975) fusion (CCDC6 exon 1 fused to RET exons 12-20): c.304-20462:CCDC6_c.2136+447:RETinv Protein Fusion: in frame {CCDC6:RET}	NA
EVT-P-002 9315-T01-I M6-146	P-0029 315-T0 1-IM6		RET- CCD C6	CC DC6	in-fr ame	thr_ee _pr im_e	436106 31	11	712	True	retain ed	1.0	False	Protei n kinase domai n	PF0 002 8		CCDC6 (NM_005436) - RET (NM_020975) fusion (CCDC6 exon 1 fused to RET exons 12-20: c.304-20462:CCDC6_c.2136+447:RETinv Protein Fusion: in frame {CCDC6:RET}	NA
EVT-P-000 0889-T01-I M3-147	P-0000 889-T0 1-IM3		RET- CCD C6	CC DC6	in-fr ame	five _pr im_e	436119 94	12	713	True	lost	0.0	False	PF000 28	Prot ein kina se d oma in		RET (NM_020630) - CCDC6 (NM_005436) Reciprocal Inversion: c.2137-38_c.2137-38inv Protein fusion: in frame (RET-CCDC6)	NA
EVT-P-000 7322-T02-I M6-148	P-0007 322-T0 2-IM6		RET- CCD C6	CC DC6	in-fr ame	thr_ee _pr im_e	436102 68	11	712	True	retain ed	1.0	False	Protei n kinase domai n	PF0 002 8		CCDC6 (NM_005436) - RET (NM_020975) fusion: c.454-1512:CCDC6_c.2136+84:RETinv Protein Fusion: in frame {CCDC6:RET}	NA
EVT-P-002 7953-T01-I M6-149	P-0027 953-T0 1-IM6		RET- CCD C6	CC DC6	in-fr ame	thr_ee _pr im_e	436112 78	12	713	True	retain ed	1.0	False	Protei n kinase domai n	PF0 002 8		CCDC6 (NM_005436) - RET (NM_020975) fusion (CCDC6 exon 1 fused to RET exons 12-20): c.303+20901:CCDC6_c.2137-754:RETinv Protein Fusion: in frame {CCDC6:RET}	NA
EVT-P-000 8504-T03-I M6-150	P-0008 504-T0 3-IM6		RET- CCD C6	CC DC6	in-fr ame	thr_ee _pr im_e	436105 83	11	712	True	retain ed	1.0	False	Protei n kinase domai n	PF0 002 8		CCDC6 (NM_005436) - RET (NM_020975) fusion (CCDC6 exons 1-8 fused to RET exons 12-20): c.1230+499:CCDC6_c.2136+399:RETinv Protein Fusion: in frame {CCDC6:RET}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intron_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site_2_effect_on_frame
EVT-P-0008955-T03-I M5-151	P-0008955-T03-IM5		RET-CCDC6	CCDC6	in-frame	threesome_prime	43611019	11	712	True	retained	1.0	False	Protein kinase domain	PF00028		CCDC6 (NM_005436) - RET (NM_020975) fusion: c.303+26028:CCDC6_c.2136+835:RETinv Protein fusion: in frame (CCDC6-RET)	NA
EVT-P-0013023-T01-I M5-152	P-0013023-T01-IM5		RET-CCDC6	CCDC6	in-frame	threesome_prime	43611810	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		CCDC6 (NM_005436) - RET (NM_020975) fusion: c.304-17904:CCDC6_c.2137-222:RETinv Protein fusion: in frame (CCDC6-RET)	NA
EVT-P-0013394-T01-I M5-153	P-0013394-T01-IM5		RET-CCDC6	CCDC6	in-frame	threesome_prime	43610232	11	712	True	retained	1.0	False	Protein kinase domain	PF00028		CCDC6 (NM_005436) - RET (NM_020975) fusion: c.303+18884:CCDC6_c.2136+48:RETinv Protein fusion: in frame (CCDC6-RET)	NA
EVT-P-0022607-T01-I M6-154	P-0022607-T01-IM6		RET-CCDC6	CCDC6	in-frame	threesome_prime	43611212	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		CCDC6 (NM_005436) - RET (NM_020975) fusion (CCDC6 exon 1 fused to RET exons 12 - 20): c.304-414:CCDC6_c.2137-820:RETinv Protein Fusion: in frame {CCDC6:RET}	NA
EVT-P-0022642-T01-I M6-155	P-0022642-T01-IM6		RET-CCDC6	CCDC6	in-frame	threesome_prime	43611768	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		CCDC6(NM_005436) - RET(NM_020975) Fusion (CCDC6 exon1 fused to RET exon12) : c.304-14214:CCDC6_c.2137-273:RETinv Protein Fusion: in frame {CCDC6:RET}	NA
EVT-P-0022642-T02-I M6-156	P-0022642-T02-IM6		RET-CCDC6	CCDC6	in-frame	threesome_prime	43611768	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		CCDC6 (NM_005436) - RET (NM_020975) Fusion (CCDC6 exon1 fused to RET exons 12-20) : c.304-14214:CCDC6_c.2137-273:RETinv Protein Fusion: in frame {CCDC6:RET}	NA
EVT-P-0017365-T03-I M6-157	P-0017365-T03-IM6		RET-CCDC6	CCDC6	in-frame	threesome_prime	43611464	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		CCDC6 (NM_005436) - RET (NM_020975) fusion (CCDC6 exon 1 fused to RET exons 12-20): c.303+25501:CCDC6_c.2137-568:RETinv Protein Fusion: in frame {CCDC6:RET}	NA
EVT-P-0025349-T03-I M6-158	P-0025349-T03-IM6		RET-CCDC6	CCDC6	in-frame	threesome_prime	43611986	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		CCDC6 (NM_005436) - RET (NM_020975) fusion (CCDC6 exon 1 fused to RET exons 12-20): c.304-10038:CCDC6_c.2137-46:RETinv Protein Fusion: in frame {CCDC6:RET}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site_2_effect_on_frame
EVT-P-004 9588-T01-I M6-159	P-0049 588-T0 1-IM6		RET- CCD C6	CC DC6	out- of-fr ame	five _pr ime	436094 37	10	627	True	lost	0.0	False	PF000 28	Protein kinase domain		RET (NM_020975) - CCDC6 (NM_005436) rearrangement: c.1879+314:RET_c.304-1522 2:CCDC6inv Protein Fusion: out of frame {RET:CCDC6}	NA
EVT-P-001 5435-T01-I M6-160	P-0015 435-T0 1-IM6		RET- CCD C6	CC DC6	in-fr ame	three _pr ime	436109 69	11	712	True	retained	1.0	False	Protein kinase domain	PF0002 8		CCDC6 (NM_005436) - RET (NM_020975) fusion (CCDC6 exon 1 fused to RET exon 12): c.304-21366:CCDC6_c.2136+785:RETinv Protein fusion: in frame (CCDC6-RET)	NA
EVT-P-000 5556-T01-I M5-161	P-0005 556-T0 1-IM5		RET- CCD C6	CC DC6	in-fr ame	three _pr ime	436118 28	12	713	True	retained	1.0	False	Protein kinase domain	PF0002 8		CCDC6 (NM_005436) - RET (NM_020975) fusion (CCDC6 exon1 with RET exons 12-20) - c.303+21888:CCDC6_c.2137-204:RETinv Protein fusion: in frame (CCDC6-RET)	NA
EVT-P-006 3502-T01-I M7-162	P-0063 502-T0 1-IM7		RET- CTN NA1	CTN NA1	unknown	five _pr ime	436220 38	19	1019	False	retained	1.0	False	Protein kinase domain; PF00028			RET (NM_020975) - CTNNA1 (NM_001903) fusion: t(5;10)(q31.2;q11.21)(chr5:g.138223827::chr10:g.43622038) Protein Fusion: mid-exon {RET:CTNNA1}	NA
EVT-P-006 3802-T01-I M7-163	P-0063 802-T0 1-IM7		RET- CTN NA3	CTN NA3	out- of-fr ame	five _pr ime	436175 66	16	934	True	disrupted	0.7482269 5035461	True	PF000 28		Protein kinase domain	RET (NM_020975) - CTNNA3 (NM_013266) rearrangement: c.2801+102:RET_c.1375-342 98:CTNNA3inv Protein Fusion: out of frame {RET:CTNNA3}	NA
EVT-P-003 0396-T02-I M6-164	P-0030 396-T0 2-IM6		RET- DOC K1	DO CK1	unknown	three _pr ime	436083 20	9	556	False	retained	1.0	False	Protein kinase domain	PF0002 8		DOCK1 (NM_001380) - RET (NM_020975)Rearrangement: c.2515+295:RET_c.1668:RETdup Protein Fusion: mid-exon {DOCK1:RET}	NA
EVT-P-001 3886-T01-I M5-165	P-0013 886-T0 1-IM5		RET- EML 4	EML 4	unknown	three _pr ime	436100 67	11	673	False	retained	1.0	False	Protein kinase domain	PF0002 8		EML4 (NM_019063) - RET (NM_020975) rearrangement: t(2;10)(p21;q11.21)(chr2:g.42550559::chr10:g.43610067) Protein fusion: mid-exon (EML4-RET)	NA
EVT-P-001 4808-T03-I M6-166	P-0014 808-T0 3-IM6		RET- ERC C6	ERC C6	out- of-fr ame	five _pr ime	436112 16	12	713	True	lost	0.0	False	PF000 28	Protein kinase domain		RET (NM_020975) - ERCC6 (NM_000124) rearrangement: c.2137-816:RET_c.1397+682 2:ERCC6inv Protein Fusion: out of frame {RET:ERCC6}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-001 4808-T02-I M6-167	P-0014 808-T0 2-IM6		RET- ERC C6	ERC C6	out- of-fr ame	five _pr ime	436112 16	12	713	True	lost	0.0	False	PF000 28	Prot ein kina se d oma in		RET (NM_020975) - ERCC6 (NM_000124) rearrangement: c.2137-816:RET_c.1397+682 2:ERCC6inv Protein Fusion: out of frame {RET:ERCC6}	NA
EVT-P-001 4808-T01-I M6-168	P-0014 808-T0 1-IM6		RET- ERC C6	ERC C6	out- of-fr ame	five _pr ime	436112 16	12	713	True	lost	0.0	False	PF000 28	Prot ein kina se d oma in		RET (NM_020975) - ERCC6 (NM_000124) rearrangement: c.2137-816:RET_c.1397+682 2:ERCC6inv Protein fusion: out of frame (RET-ERCC6)	NA
EVT-P-004 5809-T02-I M6-169	P-0045 809-T0 2-IM6		RET- FBXL 7	FBX L7	in-fr ame	five _pr ime	436222 69	19	1063	True	retai ned	1.0	False	Protei n kinase domai n; PF0 0028			RET (NM_020975) - FBXL7 (NM_012304) fusion: t(5;10)(p15.1;q11.21)(chr5:g.15556734::chr10:g.43622269) Protein Fusion: in frame {RET:FBXL7}	NA
EVT-P-004 5809-T06-I M6-170	P-0045 809-T0 6-IM6		RET- FBXL 7	FBX L7	in-fr ame	five _pr ime	436223 23	19	1063	True	retai ned	1.0	False	Protei n kinase domai n; PF0 0028			RET (NM_020975) - FBXL7 (NM_012304) fusion: t(5;10)(p15.1;q11.21)(chr5:g.15556734::chr10:g.43622323) Protein Fusion: in frame {RET:FBXL7}	NA
EVT-P-002 5215-T01-I M6-171	P-0025 215-T0 1-IM6		RET- GEMI N5	GE MIN 5	in-fr ame	thre e _pr ime	436110 67	11	712	True	retai ned	1.0	False	Protei n kinase domai n	PF0 002 8		GEMIN5 (NM_015465) - RET (NM_020975) rearrangement: t(5;10)(q33.2;q11.21)(chr5:g.154274993::chr10:g.43611067) Protein Fusion: in frame {GEMIN5:RET}	NA
EVT-P-002 7417-T01-I M6-173	P-0027 417-T0 1-IM6		RET- KCN MA1	KCN MA1	out- of-fr ame	thre e _pr ime	436081 58	9	550	True	retai ned	1.0	False	Protei n kinase domai n	PF0 002 8		KCNMA1 (NM_001161352) - RET (NM_020975) rearrangement: c.378+15627: KCNMA1_c.1649-143:RET Protein Fusion: out of frame {KCNMA1:RET}	NA
EVT-P-004 3704-T01-I M6-174	P-0043 704-T0 1-IM6		RET- KIF5 B	KIF 5B	in-fr ame	five _pr ime	436103 31	11	712	True	lost	0.0	False	PF000 28	Prot ein kina se d oma in		RET (NM_020975) - KIF5B (NM_004521) rearrangement: c.2136+147:RET_c.2545-77:KIF5Binv Protein Fusion: in frame {RET:KIF5B}	NA
EVT-P-000 6226-T02-I M5-178	P-0006 226-T0 2-IM5		RET- NCO A4	NC OA4	in-fr ame	thre e _pr ime	436118 93	12	713	True	retai ned	1.0	False	Protei n kinase domai n	PF0 002 8		NCOA4 (NM_001145260) - RET (NM_020975) fusion : c.762+254:NCOA4_c.2137-139: RETdup Protein fusion: in frame (NCOA4-RET)	NA
EVT-P-000 1151-T01-I M3-179	P-0001 151-T0 1-IM3		RET- NCO A4	NC OA4	in-fr ame	thre e _pr ime	436116 45	12	713	True	retai ned	1.0	False	Protei n kinase domai n	PF0 002 8		RET (NM_020630) - NCOA4 (NM_005437) Duplication : c.2137-387_c.1840-75dup Protein fusion: in frame (NCOA4-RET)	NA

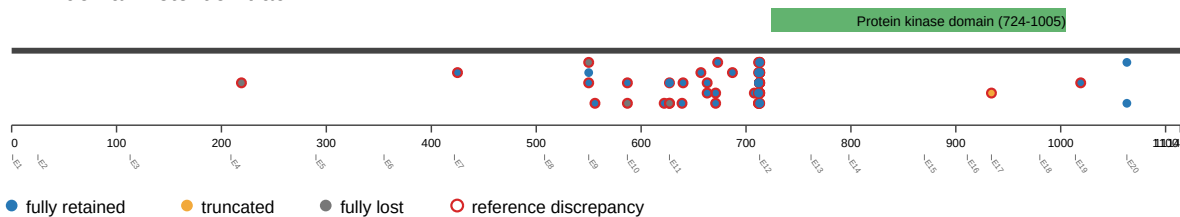
event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-000 9592-T01-I M5-180	P-0009 592-T0 1-IM5		RET- NCO A4	NC OA4	unknown	thre e_ pr ime	436111 43	12	713	True	retained	1.0	False	Protein kinase domain	PF0 002 8		NCOA4 (NM_001145260) - RET (NM_020975) fusion (NCOA4 exons 1-9 with RET exons 12=20: c.872:NCOA4_c .2136+621:RETdup Protein fusion: mid-exon (NCOA4-RET)	NA
EVT-P-000 1155-T01-I M3-181	P-0001 155-T0 1-IM3		RET- NCO A4	NC OA4	in-frame	thre e_ pr ime	436109 41	11	712	True	retained	1.0	False	Protein kinase domain	PF0 002 8		RET (NM_020630) - NCOA4 (NM_005437) Duplication : c.2136+757_c.715-617dup Protein fusion: in frame (NCOA4-RET)	NA
EVT-P-002 2818-T01-I M6-182	P-0022 818-T0 1-IM6		RET- NCO A4	NC OA4	unknown	thre e_ pr ime	436117 46	12	713	True	retained	1.0	False	Protein kinase domain	PF0 002 8		NCOA4 (NM_001145260) - RET (NM_020975) fusion (NCOA4 exons 1-8 fused with RET exons 12-20): c.618+42: NCOA4_c.2137-860:RETdup Protein Fusion: mid-exon {NCOA4:RET}	NA
EVT-P-000 2145-T01-I M3-183	P-0002 145-T0 1-IM3		RET- PDC D10	PDC D10	out-of-frame	five _ pr ime	436093 94	10	627	True	lost	0.0	False	PF000 28	Protein kinase domain		RET (NM_020975) - PDCD10 (NM_145860) translocation: t(10;3)(q11.21;q26.1)(chr10:g.4 3609394::chr3:g.167419184) Protein fusion: out of frame (RET-PDCD10)	NA
EVT-P-000 1898-T01-I M3-184	P-0001 898-T0 1-IM3		RET- RAS SF4	RAS SF4	in-frame	five _ pr ime	436117 49	12	713	True	lost	0.0	False	PF000 28	Protein kinase domain		RET (NM_020975) - RASSF4 (NM_032023) deletion: c.2137-283_c.138+357del Protein fusion: in frame (RET-RASSF4)	NA
EVT-P-003 0396-T02-I M6-185	P-0030 396-T0 2-IM6		RET- RBP MS	RBP MS	unknown	thre e_ pr ime	436100 18	11	657	False	retained	1.0	False	Protein kinase domain	PF0 002 8		RBPMs (NM_001008712) - RET (NM_020975) Rearrangement : t(8;10)(p21.1 ,q11.21)(chr8:g.30412656::chr 10:g.43610018) Protein Fusion: mid-exon {RBPMs:RET}	NA
EVT-P-002 9222-T02-I M6-212	P-0029 222-T0 2-IM6		RET- RUF Y1	RUF Y1	unknown	thre e_ pr ime	436100 37	11	663	False	retained	1.0	False	Protein kinase domain	PF0 002 8		RUFY1 (NM_025158) - RET (NM_020975) Rearrangement : t(5;10)(q35.3,q11.21)(chr5:g. 179025890::chr10:g.4361003 7) Protein Fusion: mid-exon {RUFY1:RET}	NA
EVT-P-001 5310-T02-I M6-213	P-0015 310-T0 2-IM6		RET- RUF Y3	RUF Y3	out-of-frame	thre e_ pr ime	436104 27	11	712	True	retained	1.0	False	Protein kinase domain	PF0 002 8		RUFY3 (NM_001037442) - RET (NM_020975) rearrangement: t(4;10)(q13.3; q11.21)(chr4:g.71655316::chr 10:g.43610427) Protein fusion: out of frame (RUFY3-RET)	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site_2_effect_on_frame
EVT-P-001 5310-T04-I M6-214	P-0015 310-T0 4-IM6		RET-RUF Y3	RUF Y3	unknown	threep _prime	436104 27	11	712	True	retained	1.0	False	Protein kinase domain	PF0 002 8		RUFY3 (NM_001037442) - RET (NM_020975) rearrangement: t(4;10)(q13.3; q11.21)(chr4:g.71655251::chr 10:g.43610427) Protein Fusion: mid-exon {RUFY3:RET}	NA
EVT-P-002 5609-T01-I M6-215	P-0025 609-T0 1-IM6		RET-SH ROOM3	SHR OOM3	in-frame	threep _prime	436115 45	12	713	True	retained	1.0	False	Protein kinase domain	PF0 002 8		SHROOM3 (NM_020859) - RET (NM_020975) rearrangement: t(4;10)(q21.1; q11.21)(chr4:g.77664882::chr 10:g.43611545) Protein Fusion: in frame {SHROOM3:RET}	NA
EVT-P-005 3530-T01-I M6-216	P-0053 530-T0 1-IM6		RET-SLC 4A4	SLC 4A4	out-of-frame	fivep _prime	436104 39	11	712	True	lost	0.0	False	PF000 28	Protein kinase domain		RET (NM_020975) - SLC4A4 (NM_001134742) rearrangement: t(4;10)(q13.3; q11.21)(chr4:g.72412310::chr 10:g.43610439) Protein Fusion: out of frame {RET:SLC4A4}	NA
EVT-P-000 2145-T01-I M3-217	P-0002 145-T0 1-IM3		RET-TFG	TFG	in-frame	threep _prime	436094 37	10	627	True	retained	1.0	False	Protein kinase domain	PF0 002 8		RET (NM_020975) - TFG (NM_001195478) translocation: t(10;3)(q11.21;q 12.2)(chr10:g.43609437::chr3: g.100450751) Protein fusion: in frame (TFG-RET)	NA
EVT-P-002 1980-T01-I M6-218	P-0021 980-T0 1-IM6		RET-TIM M23B	TIM M23B	out-of-frame	threep _prime	436105 50	11	712	True	retained	1.0	False	Protein kinase domain	PF0 002 8		NCOA4 (NM_001145260) - RET (NM_020975) fusion (NCOA4 exons 1-8 fused to RET exons 12- 20): c.763-591 :NCOA4_c.2136+366:RETdup Protein Fusion: out of frame {TIMM23B:RET}	NA
EVT-P-001 7716-T01-I M6-219	P-0017 716-T0 1-IM6		RET-TIM M23B	TIM M23B	out-of-frame	threep _prime	436111 72	12	713	True	retained	1.0	False	Protein kinase domain	PF0 002 8		NCOA4 (NM_001145260) - RET (NM_020975) fusion (NCOA4 exons 1-7 fused with RET exons 12-20): c.618+42: NCOA4_c.2137-860:RETdup Protein Fusion: out of frame {TIMM23B:RET}	NA
EVT-P-002 5531-T01-I M6-220	P-0025 531-T0 1-IM6		RET-TIM M23B	TIM M23B	out-of-frame	threep _prime	436113 75	12	713	True	retained	1.0	False	Protein kinase domain	PF0 002 8		NCOA4 (NM_001145260) RET (NM_020975) fusion: c.6 19-108NCOA4_c.2137-657RE Tdup Protein Fusion: out of frame {TIMM23B:RET}	NA
EVT-P-003 4704-T01-I M6-222	P-0034 704-T0 1-IM6		RET-TRIM 24	TRI M24	in-frame	threep _prime	436118 27	12	713	True	retained	1.0	False	Protein kinase domain	PF0 002 8		TRIM24 (NM_015905) - RET (NM_020975) Fusion (TRIM24 exon 9 fused with RET exon12) : t(7;10)(q34;q11.21)(chr7:g.138243011::chr10:g.43 611827) Protein Fusion: in frame {TRIM24:RET}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site_2_effect_on_frame
EVT-P-004 6790-T01-I M6-223	P-0046 790-T0 1-IM6		RET- UXS 1	UXS 1	unknown	five_prime	43600430	4	219	False	lost	0.0	False		Protein kinase domain	PF00028	RET (NM_020975) - UXS1 (NM_001253875) rearrangement: t(2;10)(q12.2;q11.21)(chr2:g.106728260::chr10:g.43600430) Protein Fusion: mid-exon {RET:UXS1}	NA
EVT-P-002 9222-T03-I M6-224	P-0029 222-T0 3-IM6		RUF Y1-R ET	RUF Y1	unknown	three_prime	43610037	11	663	False	retained	1.0	False	Protein kinase domain	PF00028		RUFY1 (NM_025158) - RET (NM_020975) fusion: t(5;10)(q35.3;q11.21)(chr5:g.179025890::chr10:g.43610037) Protein Fusion: mid-exon {RUFY1:RET}	NA
EVT-P-003 1340-T02-I M6-225	P-0031 340-T0 2-IM6		RUF Y2-R ET	RUF Y2	in-frame	three_prime	43611559	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		RUFY2 (NM_017987) - RET (NM_020975) fusion: c.1045-640:RUFY2_c.2137-473:RETinv Protein Fusion: in frame {RUFY2:RET}	NA
EVT-P-002 6614-T01-I M6-227	P-0026 614-T0 1-IM6		SPE CC1L -RET	SPE CC1 L	in-frame	three_prime	43611575	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		SPECC1L (NM_001145468) - RET (NM_020975) fusion (SPECC1L exons 1-10 fused to RET exons 12-20): t(10;22)(10q11.21;22q11.23)(chr10:g.43611575::chr22:g.24740413) Protein Fusion: in frame {SPECC1L:RET}	NA
EVT-P-003 6400-T06-I M6-228	P-0036 400-T0 6-IM6		SPE CC1L -RET	SPE CC1 L	in-frame	three_prime	43610353	11	712	True	retained	1.0	False	Protein kinase domain	PF00028		SPECC1L (NM_001145468) - RET (NM_020975) fusion: t(10;22)(q11.21;q11.23)(chr10:g.43610353::chr22:g.24739056) Protein Fusion: in frame {SPECC1L:RET}	NA
EVT-P-003 6400-T04-I M6-229	P-0036 400-T0 4-IM6		SPE CC1L -RET	SPE CC1 L	in-frame	three_prime	43610353	11	712	True	retained	1.0	False	Protein kinase domain	PF00028		SPECC1L (NM_001145468) - RET (NM_020975) fusion (SPECC1L exons 1-9 fused to RET exons 12-20): t(10;22)(q11.21;q11.23)(chr10:g.43610353::chr22:g.24739056) Protein Fusion: in frame {SPECC1L:RET}	NA
EVT-P-003 6400-T03-I M6-230	P-0036 400-T0 3-IM6		SPE CC1L -RET	SPE CC1 L	in-frame	three_prime	43610353	11	712	True	retained	1.0	False	Protein kinase domain	PF00028		SPECC1L (NM_001145468) - RET (NM_020975) fusion (SPECC1L exons 1-9 fused to RET exons 12-20): t(10;22)(q11.21;q11.23)(chr10:g.43610353::chr22:g.24739056) Protein Fusion: in frame {SPECC1L:RET}	NA

domain retention outliers

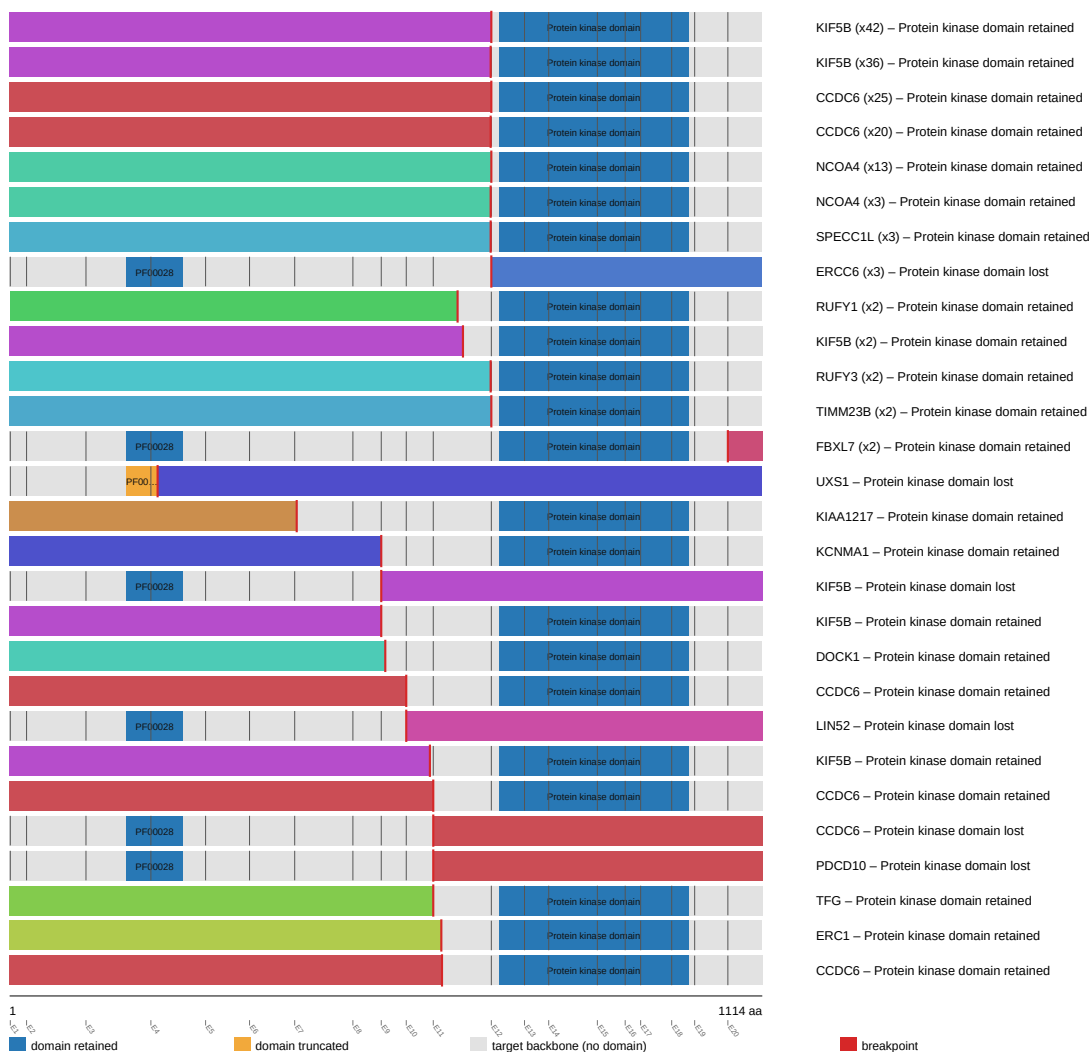
RET domain-retention track



fusion schematic

RET fusion-transcript schematic

partner block → breakpoint → retained RET portion (domain-colored, exon ticks)



Showing the top 28 of 52 partner/breakpoint groups by recurrence; see results.tsv for the complete list.

reference comparison

Reference vs msk_impact_50k_2026

